Lecture 41 — Capstone studio

Mon Dec 7

Last updated: October 8, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Studio day for the capstone (no new mathematics lecture). Related objectives: Lectures. Schedule.

Prepare before class

1. Bring a runnable draft of Plan A vs Plan B from the capstone / Lab 7 launch (Lecture 40).
2. Know which error, storage, and runtime quantities you plan to report (see capstone success criteria).
3. Bring the Lab 7 ticket if you still need studio initials.

Learning objectives

By the end of class, students should be able to:

1. Implement and validate a low-rank computational strategy.
2. Measure approximation error, storage, and runtime.
3. Separate one-time factorization cost from repeated-use cost.
4. Compare theoretical operation/storage counts with empirical measurements.
5. Debug and refine an analysis collaboratively while retaining individual understanding.

Agenda

- Brief check-in: question, matrix/data choice, and planned rank(s).
- Studio block: implement / validate, log error–storage–runtime numbers.
- Instructor / peer debugging checkpoints (keep individual ownership of the analysis).
- Exit ticket: what you will show on Wednesday's gallery and what still needs a number.